

1).

i) * When the two plastic rods are rubbed with wool, both rods gain the same type of charge (negative charge)

* According to the property of electric charges:

- Like charges repel each other
- Unlike charges attract each other.

* $\therefore$ when the rods are brought close together, they move away from each other due to electrostatic repulsion.

2. The basic classification of electric charges is

- 1) Positive charge (+) - Produced when an object loses electrons.
- 2) Negative charge (-) - Produced when an object gains electrons

3. * According to law of conservation of electric charge.

It says that, "Electric charge can neither be created nor destroyed. It can only be transferred from one body to another"

* When the charged sphere touches the neutral sphere, the charge redistributes between them. Although the amount of charges on each sphere changes. The total charge of the system remains constant and equals the original charge ~~be~~ present before contact.

4. * The physical law illustrated is the law of conservation of electric charges:

* Electric charge can neither be created nor destroyed; it can only be transferred from one body to another.

5. * Electric charge is responsible for producing electrostatic forces between objects

* When the plastic comb is rubbed with dry hair, it becomes electrically ^{charged} due to friction. The charged comb creates an electric field that attracts the neutral paper pieces by inducing opposite charges in them.

* As a result, an electrostatic force acts between the comb and the paper pieces, causing them to move toward the comb and stick to it.

6. * ~~When~~ The physical property responsible for this electrostatic interaction is electric charge.

* When the plastic scale is rubbed with a dry cloth, it becomes electrically charged. This charge produces electrostatic forces that attract the nearby thermocol and paper ~~the~~ pieces.

7. * This demonstrates the law of conservation of electric charge, which states that: "Electric charge can ~~either~~ neither be created nor destroyed, only transferred from one body to another."

* When the two metal ^{plates} are connected by a conducting wire, electric charge flows from one plate to the other until both reach equilibrium.

* During this transfer, no charge is created or destroyed; it is redistributed between the plates.

8. * This states the law of conservation of electric charge.
* "Electric charge can neither be created nor destroyed, only transferred from one body to another."

9. * Quantization of electric charges means electric charge exists only in discrete amounts and not in ~~arbit~~ arbitrary values.

* The charge on a body is given by:
 $q = ne$ [$\therefore$ where $e = 1.6 \times 10^{-19} \text{ C}$ and n is integers]

significance in Electrostatics:

- * It proves that electric charge is discrete in nature.
- * It helps explain charging by gain or loss of electrons.
- * It forms the basis for understanding atomic structure and electrostatic phenomena.

10. * Metal contains large number of free electrons.

* In the absence of a battery, electrons move

randomly.

* When a battery is connected, an electric field is produced in wire.

* Electron drift slowly in a particular direction due to this field; this is called electron drift.

* The directed drift of electrons causes the flow of electric current in metallic conductors.

11. Principle of Additivity of Electric charges:

- * Electric charge is an algebraic quantity with positive and negative signs.

- * The total charge of a system is the sum of all individual charge.

$$Q_{\text{total}} = Q_1 + Q_2 + Q_3 + \dots$$

- * When charges are combined, we simply add them algebraically.

- * Positive and negative charge can cancel each other during addition.

- * Hence, the total charge remains conserved even after interaction of bodies.

12. * Conductors such as copper and aluminium, contain a large number of free electrons in their outer shells, and

- * These free electrons are not tightly bound to atoms, can move easily inside the material.

- * When a voltage is applied, an electric field is created inside the conductor.

- * The free electrons start moving in a particular direction under the influences of this electric field, producing electric current.

- * Since, conductors have many mobile electrons and low resistance, they allow electric charges to flow efficiently and carry current easily.

13. * The law of conservation of electric charge states that Electric charge can neither be created nor destroyed, only transferred from one body to another.

* When the charged and neutral spheres are connected, electron flow through the wire.
* Charge redistribute until both spheres reach equilibrium.

* The amount of charge on each sphere changes after sharing.

* However, the total charge of the system remains constant proving conservation of charge.

14. * Inside thunderclouds, ice crystals and water droplets collide due to strong air currents.

* During these collisions, electrons are transferred from one particle to another.

* As a result, some particles become positively charged while others become negatively charged.

* Lighter positive charges move to the upper part of the cloud, while heavier negative charges collect at the lower part.

* This separation of charges creates a large potential difference, leading to atmospheric ~~electricity~~ electricity and lightning.

15. * In particle interactions, electrons and protons may exchange or transfer charge during collisions.

* New particles may also be produced in interaction.

* However, the total electric charges before interaction is always equal to the total charge after the interaction.

* Positive and negative charges balance each other during particle creation or destruction.

* These observation proves the law of conservation of electric charge, which states the charge cannot be created or destroyed.

16. Same as 14th Qn

17. * According to the principle of superposition, the resultant electric field at a point is the vector sum of the electric fields produced by all charges.

* Each charge contributes its own magnitude and direction to the total Electric field.

* When one charge is removed, its field contribution disappears, so the resultant electric field changes in magnitude and direction.

18. * The electric field around a positive charge is directed outward from the charge.

* Hence, a positive test charge is repelled and moves away from the source charge due to repulsive force.

19. Given: $Q = 8 \mu\text{C} = 8 \times 10^{-6} \text{ C}$, $A = 2 \text{ m}^2$

$$\sigma = \frac{Q}{A}$$

$$\sigma = \frac{8 \times 10^{-6}}{2} = 4 \times 10^{-6} \text{ Cm}^{-2}$$

surface charge density $= 4 \times 10^{-6} \text{ Cm}^{-2}$

a0. * Linear charge density is used because the charge is distributed along the length of the rod.
* It represents the amount of charge per unit length of the conductor.

a1. * The electric field due to a point charge varies inversely with the square of the distance

$$E \propto \frac{1}{r^2} \quad \therefore \frac{E_1}{E_2} = \frac{r_2^2}{r_1^2}$$

Given: $E_1 = 12000 \text{ NC}^{-1}$, $r_1 = 1 \text{ m}$, $E_2 = ?$, $r_2 = 2 \text{ m}$

$$\frac{E_1}{E_2} = \frac{r_2^2}{r_1^2} \Rightarrow \frac{12000}{E_2} = \frac{2^2}{1^2}$$

$$E_2 = \frac{12000}{4} \Rightarrow \boxed{E = 3000 \text{ NC}^{-1}} \text{ New electric field at } 2 \text{ m.}$$

a2. * Two equal positive charges produce equal and opposite fields at the midpoint, so the net field is zero.

* A third charge adds another field, so the cancellation ~~not~~ no longer occurs.

a3. * In a continuous charge distribution, charge is spread over the entire wire rather than concentrated at one point.

* Each small charge element produces its own electric field at the observation point.

* The total electric field is obtained by adding the contributions of all these small elements.

24. * Real objects have charge distributed over their surfaces, length or volumes rather than at a single point.
* continuous charge distribution models help accurately calculate the electric field produced by such systems.

25. * Electric field lines move outward from positive charges and inward toward negative charge.
* Their symmetrical pattern shows the distribution of the electric field around the charge.
* The direction of lines gives the field direction and line spacing shows field strength.
* Hence, field lines help visualize the behaviour of electric fields clearly.

26. $E = \frac{\lambda}{2\pi\epsilon_0 r}$ $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$

Given:

$\lambda = 5 \times 10^{-6} \text{ C/m}$, $r = 0.5 \text{ m}$

$$E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{5 \times 10^{-6}}{2 \times 3.14 \times 8.85 \times 10^{-12} \times 0.5}$$

$E \approx 1.8 \times 10^5 \text{ NC}^{-1}$ It is the magnitude of electric field

27. * Electric field lines originate outward from positive charges.
* Since both charges are positive, they repel each other.
* $\therefore$ the field lines bend away from the region between the charges instead of crossing.

* The curved pattern shows the combined effect of the two electric fields.

* This behaviour indicates electrostatic repulsion between the two positive charges.

28. * An infinite charged wire has uniform cylindrical symmetry.

* The electric field is directed outward perpendicular to the wire.

* At equal distances from the wire, the electric field is the same everywhere.

* The field does not vary along the length of the wire.

* Hence, the electric field depends only on the perpendicular distance from the wire.

29. Given: $q = 3 \mu\text{C} = 3 \times 10^{-6} \text{ C}$

$d = 4 \text{ cm} = 4 \times 10^{-2} \text{ m}$

$$p = q \times d$$

$$p = 3 \times 10^{-6} \times 4 \times 10^{-2}$$

$$p = 12 \times 10^{-8}$$

Electric dipole moment $= 1.2 \times 10^{-7} \text{ Cm}$

30. * A uniformly charged spherical shell has perfect spherical symmetry.

* Inside the hollow cavity, electric fields from all parts of the shell cancel each other, so the net electric field is zero.

- * On the surface, the electric field reaches its maximum value because the charge is present there.
- * Outside the shell, the sphere behaves like a point charge concentrated at its center.
- * $\therefore$ the electric field outside decreases with increasing distance from the sphere.

31. * An electric dipole is formed by two equal and opposite charges separated by small distances.
- * The electric fields of the two charges combine to produce a directional field pattern.
 - * Unlike a single charge, the dipole field depends on both distance and direction around the dipole.
 - * Increasing the separation distance increases the dipole moment because $p = q \times d$.
 - * Hence, a larger separation produces a stronger and more noticeable dipole electric field.

32. $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$
 Given: $Q = 6 \times 10^{-6} \text{ C}$, $r = 1 \text{ m}$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

$$= 9 \times 10^9 \times \frac{6 \times 10^{-6}}{1^2}$$

$$= 54 \times 10^3$$

$$E = 5.4 \times 10^4 \text{ NC}^{-1}$$

Electric field magnitude.

33. * The magnetic field produced by an electromagnet is directly proportional to the current flowing through coil.

~~* When the~~ $B \propto I$

* When the current increases from 2A to 6A, more moving charges flow through the coil, producing a stronger magnetic field.

* $\therefore$ increasing current improves the strength and performance of electromagnetic devices such as electromagnets and motors.

34. * Increasing the current in the coil increases the magnetic field strength around it.

* A stronger magnetic field improves the lifting force of the electromagnetic lifting systems carry heavier loads.

35. * A moving electron in a circular ~~orbits~~ orbits behave like a tiny current loop.

* The magnetic dipole moment direction is determined using the right-hand rule for current.

* Since, electron current is opposite to electron motion, the magnetic dipole moment is perpendicular to the orbit and opposite to the direction of electron revolution.

36. Given : $r = 4 \times 10^{-11} \text{ m}$
 $I = 2 \times 10^{-3} \text{ A}$

$$\mu = I \times A \quad \text{--- (1)}$$

$$A = \pi r^2 = 3.14 \times (4 \times 10^{-11})^2$$

$$A = 5.024 \times 10^{-21} \text{ m}^2$$

$$\therefore \mu = 2 \times 10^{-3} \times 5.024 \times 10^{-21}$$

~~Magnetic dipole moment~~

$$\text{Magnetic dipole moment} = 1.0 \times 10^{-23} \text{ Am}^2$$

37. * Magnetic field lines go from the north pole to the south pole, showing field direction.

* The direction of the lines gives the direction of magnetic force.

* closer lines indicates a stronger magnetic field.

38. Given: $\Phi = 0.02 \text{ Wb}$, $A = 0.01 \text{ m}^2$

$$B = \frac{\Phi}{A} = \frac{0.02}{0.01} = 2$$

$$\text{Magnetic flux density (B)} = 2 \text{ T}$$

39. * paired electrons spin in opposite directions and produce equal and opposite magnetic moments.

* These opposite magnetic moments cancel each other.

* Unpaired electrons have uncancelled magnetic moments, so they contribute to atomic magnetism.

40. * The area of the electron orbit increases with radius ($A = \pi r^2$)

* $\therefore$ increasing the orbit radius increases the magnetic dipole moment

41. Given:

$$I = 12 \text{ A}, r = 0.03 \text{ m}, \mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$$

$$B = \frac{\mu_0 I}{2\pi r} = \frac{(4\pi \times 10^{-7}) \times 12}{2\pi \times 0.03}$$
$$= \frac{48 \times 10^{-7}}{0.06}$$

$$B = 8 \times 10^{-5} \text{ T}$$

Magnified field intensity

* The magnetic field is ~~directly~~ inversely proportional to the distance from the conductor.

* $\therefore$ reducing the distances, increases the magnetic field strength.

42. Given:

$$M = 0.5 \text{ Am}^2$$

$$r = 0.10 \text{ m}$$

$$\frac{\mu_0}{4\pi} = 10^{-7} \text{ T m/A}$$

$$B_{\text{axial}} = \frac{\mu_0}{4\pi} \times \frac{2M}{r^3}$$

$$= 10^{-7} \times \frac{1}{0.0001}$$

$$B = 10^{-4} \text{ T}$$

Magnetic field intensity along the axial line

* The magnetic field decreases rapidly with increasing distance ($B \propto 1/r^3$)

* $\therefore$ Increasing the distance greatly reduces the magnetic field strength.

43. step 1: Read / Input the values of current I and distance, r

step 2: $B = \frac{\mu_0 \times I}{2 \times \pi \times r}$

step 3: Display / Print the magnetic field intensity B

44. Given: $M = 0.4 \text{ Am}^2$, $r = 0.08 \text{ m}$, $\frac{\mu_0}{4\pi} = 10^{-7}$

$$B_{\text{equatorial}} = \left(\frac{\mu_0}{4\pi} \right) \frac{M}{r^3}$$

$$B = 10^{-7} \times \frac{0.4}{(0.08)^3} = \frac{0.4 \times 10^{-7}}{0.000512}$$

$$B = 7.81 \times 10^{-5} \text{ T}$$

~~Inter~~ Interpolation:

* The magnetic field on the equatorial line is weaker than the ^{axial} magnetic field

* For a magnetic dipole, the axial field is twice the equatorial field at the same distance

45. Given: $I = 2 \text{ A}$, $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$
Turns per meter for solenoid X , $n_x = 1500 \text{ turns/m}$
Turns per meter for solenoid Y , $n_y = 2500 \text{ turns/m}$

Formula for magnetic field inside a solenoid

$$B = \mu_0 n I$$

Magnetic field of solenoid, X :

$$B_x = 4\pi \times 10^{-7} \times 1500 \times 2 \Rightarrow B_x = 3.77 \times 10^{-3} \text{ T}$$

Magnetic field of solenoid Y:

$$B_y = 4\pi \times 10^{-7} \times 2500 \times 2$$

$$\Rightarrow B_y = 6.28 \times 10^{-3} \text{ T}$$

Conclusion:

*solenoid Y produces stronger magnetic field because it has more turns per meter.

46. Given:

Number of turns, $N = 120$

Initial magnetic flux, $\Phi_1 = 0.03 \text{ Wb}$

Final magnetic flux, $\Phi_2 = 0.01 \text{ Wb}$

time interval, $\Delta t = 0.02 \text{ s}$

using Faraday's law:

$$E = -N \left(\frac{\Delta \Phi}{\Delta t} \right)$$

$$E = -120 \left(\frac{-0.02}{0.02} \right)$$

$$E = -120(-1)$$

$$E = 120 \text{ V}$$

$$\Delta \Phi = \Phi_2 - \Phi_1$$

$$\Delta \Phi = 0.01 - 0.03$$

$$\Delta \Phi = -0.02 \text{ Wb}$$

The induced EMF in the coil is 120 V

47. Given:

Number of turns, $N = 1000$

length of solenoid, $L = 0.50 \text{ m}$

current, $I = 1.8 \text{ A}$

$$\mu_0 = 4\pi \times 10^{-7}$$

calculating the no. of turns per meter:

$$n = \frac{N}{L} = \frac{1000}{0.50} \Rightarrow n = 2000 \text{ turns/m}$$

Formula for magnetic field inside a solenoid:

$$B = \mu_0 n I$$

$$B = (4\pi \times 10^{-7}) \times 2000 \times 1.8$$

$$B = 4\pi \times 3.6 \times 10^{-4}$$

$$B = 14.4\pi \times 10^{-4}$$

$$B = 4.52 \times 10^{-3} \text{ T}$$

The magnetic field strength inside the solenoid is $4.52 \times 10^{-3} \text{ T}$

48. Given:

* Rate of change of magnetic flux:

$$\frac{d\Phi}{dt} = 0.02 \text{ Wb/s}$$

* Turns in coil X, $N_x = 150$

* Turns in coil Y, $N_y = 300$

Using Faraday's Law:

$$E = -N \left(\frac{d\Phi}{dt} \right)$$

Induced EMF in coil X:

$$E_x = -150 \times 0.02$$

$$E_x = -3 \text{ V}$$

$$\text{Magnitude} = 3 \text{ V}$$

Induced EMF in coil Y:

$$E_y = -300 \times 0.02$$

$$E_y = -6 \text{ V}$$

$$\text{Magnitude} = 6 \text{ V}$$